Feasibility of Using Major League Baseball 26-man Roster Spot on an Olympic-caliber Sprinter

Abstract

Major League Baseball (MLB) allows its teams to manage a "26-man active roster," where a team can designate 26 different players as "active." An MLB team can only use players from this roster during the course of a game. In this study, we examine the feasibility of adding an Olympic-caliber sprinter to a 26-man roster for the sole purpose of pinch running, and whether this would add value to a baseball team.

The 26-man roster is usually split up into two halves, one with 13 pitchers and one with 13 batters. Of those 13 batters, usually at least two have to play the catcher position due to the strain involved in playing the position. We examine whether a hypothetical Olympic-caliber sprinter carries more value than the non-catcher of the 13 batters who sees the least playing time (plate appearances, innings fielded, etc.).

We examine real-life base-out situations from 2021-2025 to include the MLB's automatic runner rule in full-inning games. We recreate an approximation of the value for a "13th batter" on an MLB roster, taking into account both their personal run contribution and the respite they give their teammates when they play a game. We also examine Olympic-caliber sprinters' performances at the 2024 Paris Summer Olympics and compare them to MLB players. By applying a calculated "usage rule", we can develop a rough value calculation of the Olympic-caliber sprinter for comparison to the 13th batter.

Under the model's timing, deployment, and substitution-cost assumptions, assigning a 26-man roster spot to an Olympic-caliber sprinter produces positive expected value relative to the constructed marginal position-player role.

1 Introduction

An MLB active roster has 26 players. Teams commonly carry 13 pitchers and 13 position players. We define the 13th batter as the least-used eligible non-catcher position player; this player typically receives few starts and limited plate appearances. Although there is a significant amount of nuance involved in a team's roster decisions, the general question we are trying to answer is whether an Olympic-caliber sprinter could provide more value than the 13th batter over the course of a season?

We compare two options for the same roster spot:

- **Option A** is the standard 13th batter. His value is his direct Wins above Replacement (WAR). It's important to note that WAR is an estimation, largely due to differences in defensive and baserunning value assigned to a player by different entities.
- **Option B** is an Olympic-caliber sprinter. He is incapable of batting or fielding. His value is his stolen-base value minus the cost of the player that he replaces. It is usually necessary to keep an additional player on the bench to swap into the game, as the sprinter is assumed to be entirely incapable of playing baseball.

WAR gives the number of wins that a player adds above a freely available replacement player (FanGraphs 2026c). The option with the larger season WAR has the larger modeled value. Some analysts also give the 13th batter a rest value, because a larger bench lets a manager rest his starters. This paper measures that rest value directly. It compares team batting on days with 12 batters with team batting on days with 13 batter.

To decide on the value of the sprinter, the paper must also define when it is optimal to use the sprinter.The decision rule compares an immediate steal attempt with the alternative via Win Probability Added (WPA). Win Probability Added (WPA) gives the change in win probability that one decision causes (FanGraphs 2026d). The value of inserting the Olympic-caliber runner into the game and immediately attempting to steal second is compared to against not making a substitution.

2 Background

Barring hypothetical edge cases, there is no realistic reason for a team not to use all the slots on a 26-man roster. A bench player provides value through batting, fielding, and rest. A pure pinch runner provides only baserunning value.

One important statistic to use in evaluating the pure pinch runner is Weighted Stolen Base Runs (wSB). wSB gives a run value to each stolen base and to each caught stealing (Klaassen 2013). FanGraphs publishes these run values for each season. Since the sprinter does not bat and does not field, the model only has to assign stolen-base value, which can be translated into WAR. This analysis covers the steal of second base and the steal of third base. It excludes other baserunning value, because that value becomes harder to significantly quantify. There is little data available for basepath running, such as going first-to-third or second-to-home.

The decision to run needs game context. Win expectancy is the probability that the batting team wins from a given game state. Markov models of the base-out states produce these probabilities (Bukiet et al. 1997). This paper builds win expectancy from the 2021-2025 plate appearances and uses it in the decision rule.

Our information mainly comes from three public sources. The 2024 Paris Summer Olympics results book gives the sprinter's cumulative times at each 10-meter point (Paris 2024 Organising Committee 2024). The Statcast running splits leaderboard gives MLB times across the same distances (Major League Baseball 2026b). The Statcast pop time leaderboard gives catcher throw times to second base (Major League Baseball 2026a).

3 Data and Methods

3.1 Data sample

All parts of the analysis use the 2021-2025 MLB seasons to account for the existence of the Automatic Runner rule. The period excludes the 2020 season, which had seven-inning doubleheaders as a result of the COVID pandemic. The analysis excludes postseason and spring-training games. Pitch-level data for every game come from Baseball Savant's Statcast search (Major League Baseball 2026d), and they supply the batting, baserunning, fielding, and win-expectancy inputs used throughout. The project collected an active roster for each team on each game date from the MLB Stats API using the baseballr package (Petti et al. 2024).

3.2 Selection of the 13th batter

Although MLB teams usually only have 13 batters on their 26-man roster at any given moment in time, over the course of a season teams can use upwards of 20 different batters. We use overlapping six-game windows to identify the least-used stable non-catcher position player at each point in the season.

Six games approximately represent one week of MLB play and provide a practical balance between short-window variability and roster changes accumulated over longer windows. Six games give players several batting opportunities and also limit the number of roster changes. A longer window would include more roster changes and more injuries, while a shorter one could introduce more variance. These windows overlap. The first window would hold games 1 to 6. The next window holds games 2 to 7. The overlap gives a new roster observation after each game. Our model uses them to describe roster use. A player entered the candidate pool only when he was active on all six dates of a window. This rule prevents a short stint on a roster from setting the 13th batter. The two players with the most season-to-date innings caught were excluded, because the catcher position is more gruelling to play than any other and it is practically impossible to not roster at least two.

The player with the fewest plate appearances (PA) in the window became the "13th batter". Ties used the season-to-date PA first, then the in-window playing time, then randomly. In 14% of the windows, the selected player had zero PA. We then calculated WAR for each player-game, which permits an exact six-game total. WAR is a summation of runs generated through batting, stolen-base, fielding, positional, league, and replacement runs (FanGraphs 2026c).

- Batting runs were calculated using Run Expectancy 24 (RE24) (FanGraphs 2026a), which calculates the expected run value a player generated in each specific PA.
- Stolen-base runs were calculated using wSB. This was used instead of a different statistic like Ultimate Base Running (UBR) (FanGraphs 2026b) due to differences in how baserunning value is assigned.
- Fielding runs are difficult to assign due to widespread disagreement in the best way to do so. This paper chose to use a simpler and older statistic, Zone Rating (ZR), due to unavailability and unreliability of statistics like Defensive Runs Saved (DRS), Ultimate

Zone Rating (UZR), and Outs Above Average (OAA). These statistics each have critics and proponents, which is why this paper chose a more general middle ground.
- Positional runs, league runs, and replacement runs are all typically agreed-upon values based on the position a player is fielding and the year the game was played in. In short, accounting for the fact that average defense at shortstop is worth more than a Designated Hitter (DH) and that someone hitting .300 in 2026 is rarer than in 1946.

Table 1 gives the results of the analysis, with the 13th batter usually performing slightly below replacement level.

Table 1: Observed production of the selected 13th batter

Season	Windows	WAR in 6 games	WAR per 162	PA in 6 games
2021	4708	-0.023	-0.617	6.006
2022	4710	-0.017	-0.457	5.636
2023	4710	-0.018	-0.474	5.341
2024	4708	-0.016	-0.422	4.882
2025	4710	-0.019	-0.521	4.513

We conducted a sensitivity analysis on the six-game window to see if a different length window would produce materially different results, but we found the eventual WAR benefits of the 13th batter to remain roughly the same. Due to the logic behind MLB scheduling, we decided that the six-game window made the most practical sense to continue using. Table 2 gives the comparison.

Table 2: Window-length sensitivity of the 13th-batter selection, 2021-2025

Window	PA in window	Zero-PA	WAR per 162	Match to six-game
4	2.87	25.6%	-0.471	70.5%
5	4.05	18.5%	-0.480	81.6%
6	5.28	14.3%	-0.498	100.0%
7	6.56	11.6%	-0.478	83.9%
8	7.90	9.7%	-0.450	76.7%

3.3 Comparison of 12-batter and 13-batter rosters

The value of rest is difficult to quantify, but it is still the main anectodal reason to opt against using the Olympian. The way we chose to evaluate the value of the 13th batter is to compare team batting performance during windows where 26-man rosters had 12 batters compared to when they had 13.

The sample holds 2,488 team-games with 12 batters and 13,336 team-games with 13 batters. The raw values were -0.001379 runs for each plate appearance with 12 batters and +0.001196 runs with 13 batters. The adjusted comparison gives each team-game a weight equal to its number of plate appearances and controls for the team and the season. It gives a change of +0.000907 runs for each plate appearance. The 95% confidence interval is -0.003920 to +0.005734, and the p-value is 0.704. The p-value is high, so the data do not show a reliable rest effect. The study gives the rest value a zero WAR effect.

3.4 A race model

Due to a few different reasons, it is difficult to directly translate sprinter split time into baseball split time. We create a race model to translate a given sprinter into a baseball setting. The race model uses the Jamaican sprinter Oblique Seville. The 2024 Paris Summer Olympics race report gives his cumulative times at each 10-meter point (Paris 2024 Organising Committee 2024). The official World Athletics result puts Seville eighth (last place) in the 100-meter final with a time of 9.91 seconds (World Athletics 2024). The model uses these times as a proxy for baseball speed. We also use 90-foot stolen base splits from the Statcast 90-foot running splits leaderboard (Major League Baseball 2026b).

The model fits one smooth curve to the eight published split times. One difference in how 100-meter sprint splits and stolen base splits are calculated is that 100-meter splits account for reaction time. Stolen base sprint splits start when the baserunner breaks. The official results give Seville a measured reaction time of 0.171 seconds, so we subtract that from his splits. The result is a runner time at each point on the basepath. Table 3 and Figure 1 give these inputs.

Table 3: Cumulative time to each point on the basepath (seconds)

Runner	5 ft	10 ft	20 ft	30 ft	45 ft	60 ft	75 ft	90 ft
Seville, reaction removed	0.52	0.82	1.27	1.64	2.14	2.60	3.04	3.46
MLB average, 2021-2025	0.55	0.87	1.38	1.81	2.40	2.96	3.49	4.05
MLB fastest, 2021-2025	0.50	0.78	1.24	1.64	2.18	2.69	3.17	3.66

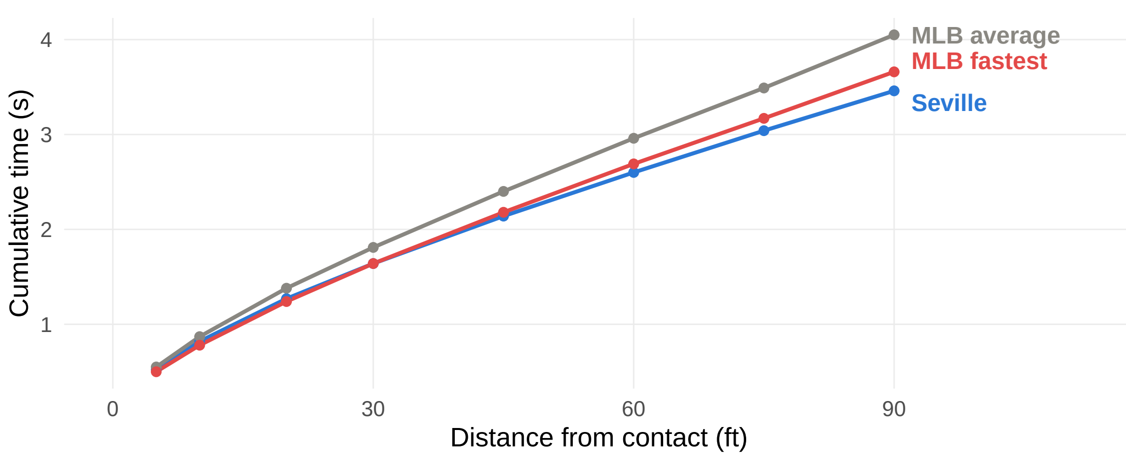

Figure 1: Cumulative time to each point on the basepath

3.5 The race-margin distribution

The race margin is the runner's arrival time at second base minus the tag's arrival time. A negative margin is safe; a margin of zero or more is out. Both clocks start when the runner breaks for second. We built the runner's clock from Seville's 2024 Paris Summer Olympics splits. Given the difference in how reaction time is measured between MLB and the Olympics, Seville's launch appears ordinary, so he takes a normal primary lead of 10.5 to 12.5 feet and gains nothing significant compared to normal MLB runners before the pitch is thrown. From a lead of (on average) 11.7 feet he has 78.3 feet to cover.

Using a kinematic model that accounts both for acceleration and slide deceleration, Seville reaches second base roughly 3.279 seconds after first movement, against 3.564 seconds for the average runner who attempts a steal. For catchers in the 2025 MLB season, the one with the lowest average pop time was J.T. Realmuto, averaging 1.86s. As a stringent catcher benchmark, we assume that Seville always faces Realmuto's 2025 mean pop time of 1.86 s.

The throw reaches second base at 3.246 seconds, marginally ahead of the runner. Seville is normally safe because we estimate that, based on stolen base attempts from 2021-2025, applying the tag takes roughly a further 0.328 seconds. All things considered, Seville will beat the tag by an average 0.295 seconds.

We model the race margin as a logistic distribution centred on the 0.295 second margin above, and we assign it a scale parameter of 0.113 seconds. This scale was found from observing tracked throws and stolen base attempts to find an estimate for variance in stolen base attempts. Under these timing and variance assumptions, the model implies a 93.1% safe rate.

3.6 Run value of one steal attempt

The value of a steal attempt by the sprinter can be given by the following formula:

$$\text{expected runs} = (\text{safe probability})(\text{steal value}) + (\text{out probability})(\text{caught-stealing value}).$$

The addition of all expected runs over the course of the season provides the value of wSB, which is the entirety of the considered positive value provided by the sprinter.

3.7 The substitution cost

MLB rules do not let the original runner return after the sprinter enters the game (Major League Baseball 2019). We estimate the direct substitution cost as the remaining offensive and defensive value of the original player minus that of his replacement. This estimate does not include the broader cost of preserving another bench player for later defensive substitution.

This substitution cost depends entirely on when the sprinter is substituted in: there will be more plate appearances left in the game if it's the sixth inning compared to the tenth inning.

We compare the run value to that of a replacement-level player per PA multiplied by the expected amount of PAs remaining.

Figure 2 gives the cost for each state in which the rule uses the sprinter. Fewer plate appearances and fewer defensive innings remain as the game goes on. The cost falls from 0.095 runs in the sixth inning to 0.009 runs in the ninth. The mean across these states is 0.043 runs, against 0.158 runs of gain from one steal attempt. The substitution takes back about half of the gain in the sixth inning, and almost none of it in the ninth.

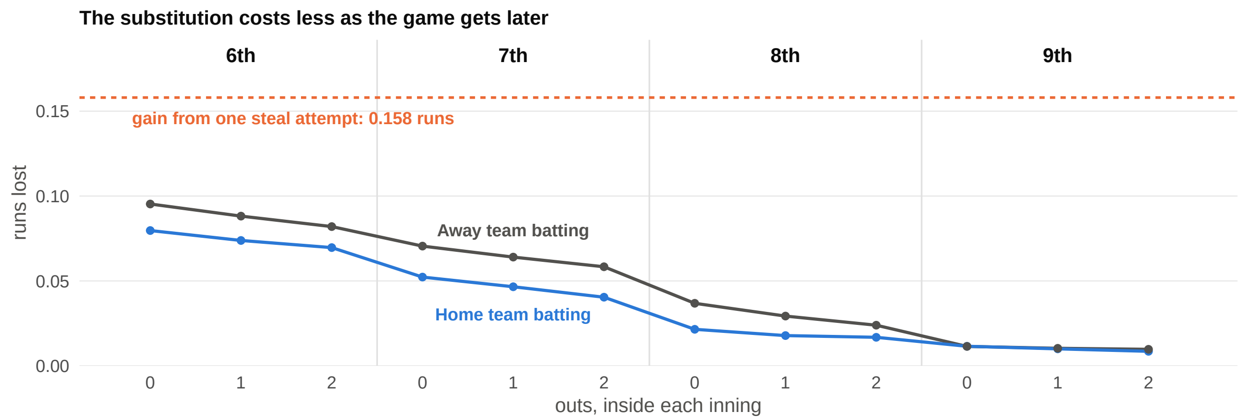

Figure 2: Run expectancy lost to the substitution, for every state in which the rule uses the sprinter.

3.8 The game-state decision rule

We examine each real plate appearance with a runner on first base and second base empty. We record the inning, the half-inning, the score, the outs, the occupied bases, the runner, the substitution cost, and the two possible results. We have to create a model that will tell us when it is "best" to use the sprinter during a game, trying to strike a chord between not substituting them in too early and not missing the best opportunity to do so.

On an immediate steal attempt after the pinch runner is substituted into the game, the model combines the win probability after a safe steal with the win probability after an out. It gives each result the weight of its modeled probability. It then subtracts the value that the team loses when the original player leaves the game. For the alternative (leaving the sprinter on the bench and using the runner on base), the model uses the observed plate-appearance results to calculate the win probability after the next opportunity. The rule is given in Table 4, which explains when to use the sprinter.

To estimate seasonal value, we applied the completed rule to each eligible 2021-2025 game state. We also used the substitution cost of its inning, half-inning, and out count. A division of the season totals by 30 teams gives the reported value for each team.

Table 4: When to use the sprinter, from the game-state model. Each cell reports the model's decision for that inning and out count; the legend below gives the run differential each symbol stands for.

Inning	0 outs	1 out	2 outs
6	–	×	×
7	✓	×	×
8	✓	✓	×
9	✓	✓	✓
10+	✓	✓	✓

Legend

✓	Use when the batting team is level, ahead, or trailing by one run (≥ -1)
–	Use only when the batting team is level or ahead (≥ 0)
×	Do not use at any run differential

3.9 Stealing third base

The sprinter is on second base after a successful steal. We apply the same model to a steal of third base. Only three inputs change: the distance, the catcher's pop time, and the run values. Seville must cover 70.1 feet. Baseball Savant records catcher throws to third base as well as to second (Major League Baseball 2026c). The archive holds 2,103 tracked throws to third base between 2021 and 2025, and 1,095 of these also carry a pop time.

The scale we calculate from that sample is 0.104 seconds, against 0.113 seconds for the throw to second base. Two separate races and two separate samples give almost the same spread, which supports the structure of the model. J.T. Realmuto averaged 1.41 seconds on his throws to third base in 2025, and we again give the sprinter that catcher on every attempt. Seville reaches third base 3.027 seconds after first movement, against 3.274 seconds for the average runner who attempts a steal. The model gives him a safe rate of 92.2%.

A steal of third base is not worth the same as a steal of second base. The gain is almost equal, but the cost of an out is much larger. The defence removes a runner who is already in scoring position. Table 5 gives the run values from the 2021-2025 RE24 table.

Table 5: Run value of a steal of third base, by out count

Outs	Runs if safe	Runs if caught	Break-even rate	Expected runs
0	+0.193	-0.873	81.9%	+0.110
1	+0.274	-0.592	68.4%	+0.206
2	+0.040	-0.333	89.3%	+0.011

The break-even rate is higher at third base, and it changes sharply with the out count. With one out it is the lowest, because a runner on third base scores on a fly ball. The sprinter is above the break-even rate in every out state.

4 Results

The standard 13th batter, on average, produced -0.498 WAR for each 162 games. The 12-player and 13-player comparison did not show a reliable rest effect, so the study adds no rest value. The total value of Option A is -0.498 WAR. This makes sense, considering the "worst" player on a roster will usually be at or around "replacement level".

Under the primary steal-of-second model, the sprinter produced 0.795 net WAR across 70.3 mean uses per season, or 1.293 WAR more than the constructed marginal position player.

The sprinter reaches second base safely on 65.5 of those uses. If he then attempts third base on each of them, he adds 8.11 runs for each team-season, which is 0.828 WAR. The combined programme is worth 1.623 WAR, or 2.121 WAR more than the 13th batter. We report the two bases separately. The every-opportunity third-base estimate is an upper-bound scenario rather than a proposed deployment rule. A steal of third base at this rate assumes an attempt frequency that no MLB runner has ever sustained.

The result comes from three separate conditions:

- The modeled safe rate is far above standard break-even rates in seasons, which are generally around 70% depending on the game-state
- The substitution cost is usually small.
- The 13th batter gives a value below replacement level in every season of the sample.

5 Case study: the 2025 Miami Marlins

The following case study is intended to illustrate the proposed deployment rule in a team-specific setting. It is not a sensitivity analysis: the simulation retains the primary model's assumptions about steal-success probability, substitution cost, and player deployment.

To test the effects of an Olympic-caliber sprinter, we applied one to the 2025 Miami Marlins. He was used in 71 game states. Using a Monte Carlo simulation, we tested the effects of the sprinter assuming the calculated stolen base rates. We scored each game with a simple points system:

- If the Olympic-caliber sprinter were to score a run in a game that the Marlins lost, and that the extra run turns into a tie, this counts as +0.5 points (to represent a 50% chance of victory).
- If the Olympic-caliber sprinter were to score a run that changes the result from a loss to a win, this counts as +1.0 point. This happens when the real-life Marlins lost in extra innings when they could have won it inside the 9 innings using the sprinter.
- The same rules apply in reverse: a win that becomes a tie counts as -0.5, and a win that becomes a loss counts as -1.0.
- All other games count as zero.

The 71 uses add roughly 21 total runs across the season. Across 400,000 simulated seasons the Marlins gain an estimated 1.34 wins and 5.05 ties. They give back 0.77 ties and 0.40 wins. The net result is 3.07 points per season under this scoring rule, and the sprinter improves the simulated record in 98.3% of seasons. Across the same 71 uses, the modeled cumulative WPA was 3.005: 1.528 from steals of second and 1.477 from steals of third.

The Marlins simulation provides an illustration of how the strategy could affect one team-season under the baseline assumptions. A complete robustness analysis would additionally vary the principal uncertain inputs, particularly the safe rate and the cost of replacing the original player after the pinch-running substitution.

6 Limitations

This analysis includes only steals of second and third base; it excludes other potential baserunning gains, including advancement on balls in play and scoring from second. It also does not model player-specific value of the incumbent runner, pitcher delivery times, pickoffs, pitchouts, or other defensive responses to an elite sprinter. The substitution estimate captures the direct remaining offensive and defensive value of the replaced player, but not the broader loss of bench flexibility when another position player must be reserved for later defensive substitution. Finally, the 12- versus 13-position-player comparison does not identify the effects of roster depth on injuries or fatigue. September's 28-player roster may increase the strategy's value, but the available sample is limited.

7 Conclusion

Using an Olympic-caliber sprinter as a dedicated pinch runner generates greater expected run value than using the 26th roster spot on a conventional position player. Under the proposed deployment rule, a steal of second base adds an estimated 1.293 wins of run value over a 162-game season. A steal of third base on each successful attempt raises this to 2.121 wins. Although this estimate depends on the runner's assumed safe rate and the game-state decision rule, simulated differences in team quality have little effect on the result. The value of the strategy therefore appears to stem primarily from the specialist's elite baserunning skill and targeted deployment, rather than from the quality of the team employing him.

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